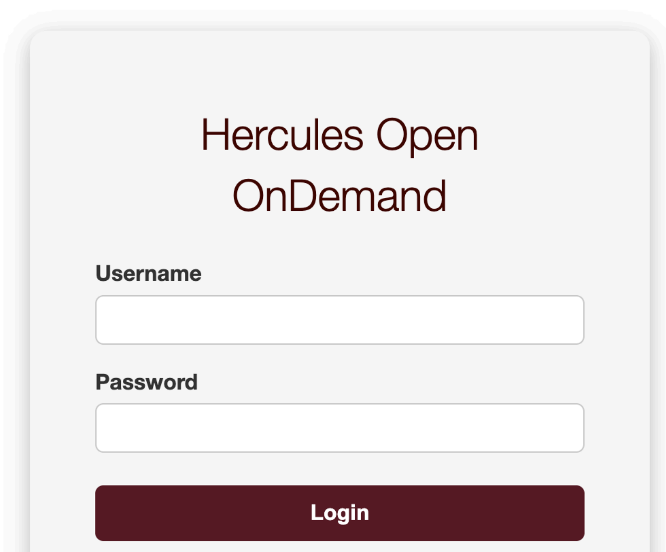
# === Hercules (NOAA/MSU) ===

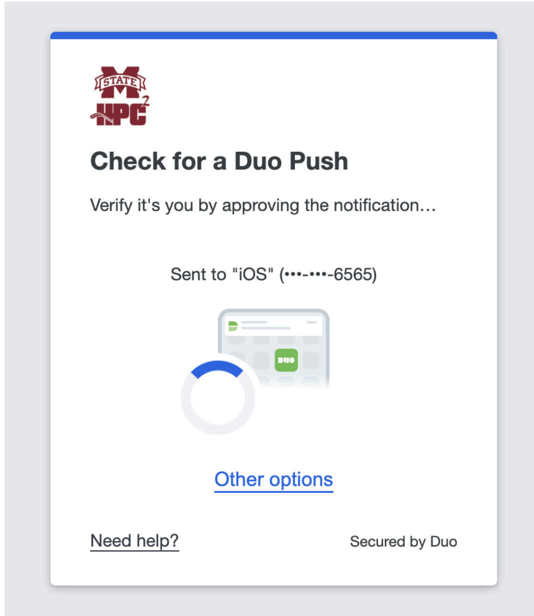
## Log-in to Hercules

1. Alternative access to Hercules:

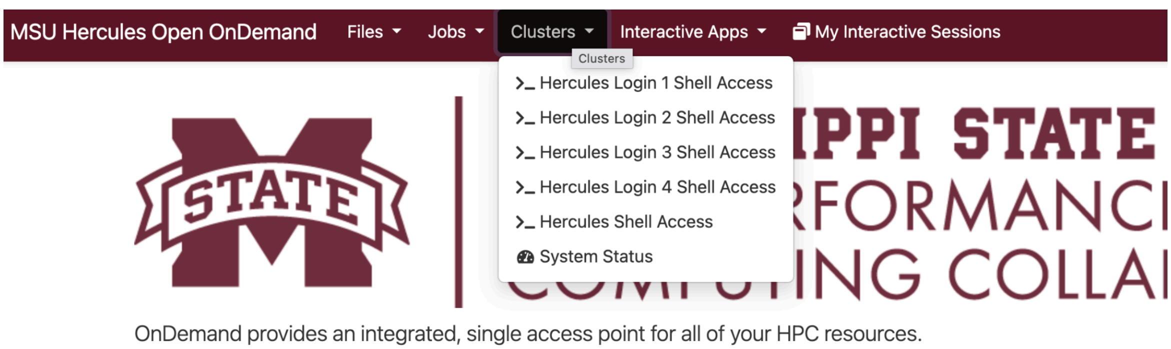
```
https://hercules-ood.hpc.msstate.edu
```

2. Log-in with your username and password:




3. Select one of the clusters:




4. Sign-in with your password and Duo:

```
********** N O T I C E **********

This system is under the control of and/or the property of Mississippi State
University (MSU).  It is for authorized use only.  By using this system, all
users acknowledge notice of and agree to comply with all MSU and High
Performance Computing Collaboratory (HPC2) policies governing use of
information systems.

Any use of this system and all files on this system may be intercepted,
monitored, recorded, copied, audited, inspected, and disclosed to authorized
university and law enforcement personnel, as well as authorized individuals of
other organizations.  By using this system, the user consents to such
interception, monitoring, recording, copying, auditing, inspection and
disclosure at the discretion of authorized university personnel.

Unauthorized, improper or negligent use of this system may result in
administrative disciplinary action, up to and including termination, civil
charges, criminal penalties, and/or other sanctions as determined by applicable
law, MSU policies, HPC2 policies, law enforcement or other authorized State
and Federal agencies.

********** N O T I C E **********

(chjeon@hercules-login-2.hpc.msstate.edu) Password:
```

5. Move to your work directory:

```
[chjeon@hercules-login-2 ~]$ pwd
/home/chjeon
[chjeon@hercules-login-2 ~]$ cd /work2/noaa/epic-explorer/chjeon/
[chjeon@hercules-login-2 chjeon]$ pwd
/work2/noaa/epic-explorer/chjeon
```

# Run JEDI (FV3-JEDI)

Note) If you have already set up the repository and input dataset on Day 1, please skip the first three steps (1 - 3).

1. Clone the following repository:

```
git clone https://github.com/NOAA-EPIC/CADRE-DA-training
```

2. Build the GDAS App (optional):

```
cd CADRE-DA-training/year2_cases

./build_gdas.sh
```

Note) The building process takes about 20 minutes on Hercules.

3. Check if the JEDI executables (‘fv3jedi_var.x’ and ‘gdas_fv3jedi_fv3inc.x’) exist in the following path (optional):

```
cd ../GDASApp

# to check the build status:

tail -n 10 build.log

# check executables in bin dir:

cd build/bin    (This is “JEDI_BIN_PATH” in the job submission script.)

ls

cd ../../../CADRE-DA-training/year2_cases
```

4. Copy the JEDI input yaml files to the specific location:

```
cd input_yaml

cp Day1/jedi_3dvar* .

(or)

cp Day2/[exp_case]/jedi_3dvar* .

(or)

cp Day3/[exp_case]/jedi_3dvar* .
```

5. Open the job submission script and modify it as needed:

```
cd ..

vim run_3dvar_hercules.sh

(Check your account, JEDI_BIN_PATH, EXP_NAME_BASE)

:wq
```

6. Submit your job:

```
sbatch run_3dvar_hercules.sh
```

7. Check your job:

```
squeue -u (your username)
```

Note) Remember the job ID of your current job.

8. Move to the experimental case directory for your job:

```
cd exp_case/[EXP_NAME_BASE].[job_ID]
```

9. Check the log files:

```
vim OUTPUT.fv3jedi

Vim errfile_fv3jedi
```

10. Move to the output directory:

```
cd output
ls
```

```
(plot_pyenv) ubuntu@ip-10-29-93-229:~/cadre26_noaa_tutorial/exp_case/cadre26.132/output$ ls
cubed_sphere_grid_atminc.jedi.nc          obs.20240224.t00z.atms_n20.satbias_cov.nc
diag.atms_n20_2024022400.nc               ufsda.t00z.atminc.cubed_sphere_grid.tile1.nc
diag.conventional_ps_2024022400.nc        ufsda.t00z.atminc.cubed_sphere_grid.tile2.nc
diag.gnssro_cosmic2_2024022400.nc         ufsda.t00z.atminc.cubed_sphere_grid.tile3.nc
diag.satwnd.abi_goes-16_2024022400.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile4.nc
diag.scatwnd.ascat_metop-b_2024022400.nc  ufsda.t00z.atminc.cubed_sphere_grid.tile5.nc
obs.20240224.t00z.atms_n20.satbias.nc     ufsda.t00z.atminc.cubed_sphere_grid.tile6.nc
```

# Run Diagnostics

1. Set up the diagnostic toolkit:

Option 1) Use the pre-installed diagnostic toolkit on Hercules:

```
source /work2/noaa/epic-explorer/cadre2026/hercules.anaconda
```

Option 2) Install the diagnostic toolkit directly:

```
cd CADRE-DA-training

git clone https://github.com/jkbk2004/ufs_da_diagnostics

# The following pip command only needs to run once

cd ufs_da_diagnostics

module load miniconda3

pip install --upgrade pip

pip install -e .

pip install seaborn scipy
```

Set up PATH in Option2:

```
cd

ls -la

cd .local/bin

cd

vim ~/.bashrc

# Adding the following line to the bottom of the file (.bashrc):

export PATH=$HOME/.local/bin:$PATH

:wq

source ~/.bashrc
```

Note) If you have set up the toolkit on Day 1, please skip the above steps.

2. Update the input YAML files for the diagnostic tools:

```
cd [your/work/directory]/CADRE-DA-training

cd diagnostic/yamls/[day(1-3)_case]

vim increment_maps.yaml

# change “prefix”s under “experiments”

:wq

vim obs_diag.yaml

# change “diag”s to the paths of your output

:wq

vim spectra_bkg_inc.yaml

# change “atm_file” under “background” and “prefix” under “increments”

:wq
```

```
vim spectra_ana_inc.yaml

# change “prefix”s under “experiments”

:wq
```

3. Run the diagnostic tools one by one:

Day1:

```
ufsda-inc-maps  --yaml increment_maps.yaml

ufsda-obs-diag  --yaml obs_diag.yaml

ufsda-spectra-bkg-inc  --yaml spectra_bkg_inc.yaml

ufsda-jedi-log [path/to/your/day1/case/directory]/OUTPUT.fv3jedi --output day1_log_report.txt
```

Day2 (nicas):

```
ufsda-inc-maps  --yaml increment_maps.yaml

ufsda-obs-diag  --yaml obs_diag.yaml

ufsda-spectra-ana-inc  --yaml spectra_ana_inc.yaml

ufsda-jedi-log [path/to/your/day2/case/directory]/OUTPUT.fv3jedi --output day2_log_report.txt
```

Day3 (error):

```
ufsda-inc-maps  --yaml ../yamls/day3_atms_err/increment_maps.yaml

ufsda-obs-diag  --yaml ../yamls/day3_atms_err/obs_diag.yaml

ufsda-spectra-ana-inc  --yaml ../yamls/day3_atms_err/spectra_ana_inc.yaml

ufsda-jedi-log [path/to/your/day3/case/directory]/OUTPUT.fv3jedi --output day3_log_report.txt
```

# Download File to Your Laptop

Open a new terminal window:

File:

```
scp [user_name]@hercules-dtn.hpc.msstate.edu:[path to the file] .
```

Directory:

```
scp -r [user_name]@hercules-dtn.hpc.msstate.edu:[path to the file] .
```